

BOUNDED CARTIER-OPERATOR DYNAMICS: UNIFORM FIBRES, FROBENIUS CORES, AND EXACT DEPTH LAWS

ANONYMOUS

ABSTRACT. Fix $q = p^a$ and bound the degree by n . We give an exact temporal and component description of the residue-class-zero Cartier section

$$\mathcal{C} \left(\sum_{j=0}^n c_j x^j \right) = \sum_{pj \leq n} c_{pj}^{1/p} x^j \quad \text{on } \mathbb{F}_q[x]_{\leq n}.$$

Every iterate is a twisted coefficient decimation, so its image has $q^{\lfloor n/p^t \rfloor + 1}$ points and every nonempty fibre has the same size $q^{n - \lfloor n/p^t \rfloor}$. An explicit $\mathbb{F}_p$ -coordinate conjugacy splits the map into inverse Frobenius and finite nilpotent shifts; each weak component is therefore a Frobenius cycle with identical attached in-trees. The first entry time τ into the constant core has exact distribution $\mathbb{P}(\tau \leq t) = q^{-\lfloor n/p^t \rfloor}$ and sharp maximum $1 + \lfloor \log_p n \rfloor$ for $n > 0$. On the core the map is inverse Frobenius: hence $\# \text{Fix}(\mathcal{C}^m) = p^{\gcd(a, m)}$, and Möbius inversion gives every cycle count and a finite Euler product for the zeta function. Along $n_L = \lfloor \alpha p^L \rfloor$, $1 \leq \alpha < p$, the reverse-depth law stabilizes exactly on every fixed lattice layer; moreover, phase size together with the fixed sequence recovers (p, a, n) . Classical Cartier extraction, finite-field Frobenius theory, and generic functional-graph machinery are treated as owned background. The residual scope is only the exact bounded specialization, lattice law, and recovery conjunction; it remains under external **HOLD**, with no novelty or priority claim.

1. INTRODUCTION AND OWNERSHIP BOUNDARY

Cartier operators convert base- p residue classes of coefficient indices into new series. In the convention of Bridy, for a perfect field of characteristic p ,

$$\Lambda_i \left(\sum_j c_j x^j \right) = \sum_j c_{pj+i}^{1/p} x^j;$$

this formula and its connection to automatic sequences are established machinery [1, Eq. (2.7)]. The terminology descends from Cartier's operation on differential forms [2]. Jeong studies two related Cartier families on finite-field power-series rings and their shift/digit extensions [5]. This note claims none of these constructions. It asks the narrower question of what exact temporal and component data result when Λ_0 is restricted to one polynomial degree bound and iterated as one map.

The answer separates into a transient coefficient forest and a periodic field automorphism. The resulting exact package is:

- (1) an explicit conjugacy to inverse Frobenius times nilpotent index-chain shifts, with the resulting product and weak-component interpretation;
- (2) a coefficient formula for every iterate, every image, and every fibre;
- (3) every core-entry shell, the sharp maximum depth, and the size of the deepest shell;
- (4) all fixed points, exact cycles, and the Artin–Mazur zeta function;
- (5) an exact lattice-depth stabilization and recovery of (p, a, n) from temporal data.

The coefficient route follows disjoint p -adic chains pointwise. A complementary $\mathbb{F}_p$ -linear reconstruction factors the iterates into coordinate restriction, inverse Frobenius, and embedding; rank-nullity then recounts the fibres and depth CDF after the iterate is known.

Owner subtraction is part of the statement. Finite-field automorphisms and their subfields are standard [6]. After restriction of scalars, the present map is an ordinary finite $\mathbb{F}_p$ -linear system. Elspas owns foundational linear state-diagram machinery [3]; Wang analyzes transition graphs of affine and linear transformations [9]; Hernández Toledo gives the general nilpotent–bijective decomposition [4]; and later functional-graph work includes Panario–Reis and Reis [7, 8]. We assign zero novelty or priority credit to the generic linear decomposition, its product of cyclic and nilpotent parts, rank/kernel counts, or

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attached-tree machinery. Likewise, Cartier/section composition, Frobenius subfields, Möbius inversion, and zeta conversion are owned inputs. Internally, the present system is not the digit-erasure map of P100, the adjugate map of P103, the annihilator–power ideal map of P107, or the nilpotent image map on a subspace lattice of P109. In particular, the phase points here are coefficient vectors rather than subspaces, and the recurrent core is the nontrivial inverse-Frobenius permutation rather than a nilpotent absorber. Rank is a proof tool, not a claim to the generic linear-engine territory. The bounded search is not an owner certificate; external circulation, novelty language, and priority language remain on **HOLD**.

2. DEFINITION AND CONVENTIONS

Let p be prime, let $a \geq 1$, set $q = p^a$, and fix $n \geq 0$. Write $\sigma(c) = c^p$ for Frobenius on $\mathbb{F}_q$. For $t \geq 0$, the notation

$$c^{p^{-t}} := \sigma^{-t}(c)$$

means the unique p^t -th root in $\mathbb{F}_q$; the superscript is automorphism notation, not a rational exponent. Thus $c^{p^{-0}} = c$, and the operation depends only on $t \bmod a$.

The phase space is

$$X_{q,n} = \mathbb{F}_q[x]_{\leq n} = \left\{ \sum_{j=0}^n c_j x^j : c_j \in \mathbb{F}_q \right\}, \quad |X_{q,n}| = q^{n+1}.$$

We keep all vectors at length $n+1$, padding high coefficients by zero, and define

$$(1) \quad \mathcal{C} \left(\sum_{j=0}^n c_j x^j \right) = \sum_{0 \leq j \leq \lfloor n/p \rfloor} c_{pj}^{p^{-1}} x^j.$$

For every $\lambda \in \mathbb{F}_q$ and $f, g \in X_{q,n}$, the defining law is

$$(2) \quad \mathcal{C}(\lambda f + g) = \sigma^{-1}(\lambda) \mathcal{C}(f) + \mathcal{C}(g), \quad \mathcal{C}^t(\lambda f + g) = \sigma^{-t}(\lambda) \mathcal{C}^t(f) + \mathcal{C}^t(g).$$

Thus $\mathcal{C}$ is additive and σ^{-1} -semilinear over $\mathbb{F}_q$; after restriction of scalars it is an ordinary $\mathbb{F}_p$ -linear endomorphism. It is generally not $\mathbb{F}_q$ -linear when $a > 1$. For the zero polynomial we set $\deg 0 = -\infty$. The constant subspace is

$$K = \mathbb{F}_q \cdot 1 \subseteq X_{q,n}.$$

The core-entry time, stable image, and periodic set are

$$(3) \quad \tau(f) = \min\{t \geq 0 : \mathcal{C}^t(f) \in K\}, \quad \text{Core}(\mathcal{C}) = \bigcap_{t \geq 0} \text{im}(\mathcal{C}^t), \quad \text{Per}(\mathcal{C}) = \{f : \mathcal{C}^m(f) = f \text{ for some } m \geq 1\}.$$

The minimum will be finite by [theorem 4.1](#). No value $v_p(0)$ and no maximum over an empty set will be used.

3. ALL ITERATES, IMAGES, AND FIBRES

The local mechanism is visible at coefficient level: only indices divisible by p survive, and every surviving index is divided by p .

Theorem 3.1 (Coefficient iterate). *For every $f = \sum_{j=0}^n c_j x^j \in X_{q,n}$ and every integer $t \geq 0$,*

$$(4) \quad \boxed{\mathcal{C}^t(f) = \sum_{0 \leq j \leq \lfloor n/p^t \rfloor} c_{p^t j}^{p^{-t}} x^j.}$$

In particular, the formula at $t = 0$ is the identity, and once $p^t > n$ the iterate is the constant $c_0^{p^{-t}}$.

Proof route I: coefficient chains. The assertion at $t = 0$ is immediate. Suppose (4) holds at time t . Applying (1) selects those displayed indices that are themselves divisible by p and applies one further inverse Frobenius:

$$\mathcal{C}^{t+1}(f) = \sum_{p^{t+1}j \leq n} \left(c_{p^t(pj)}^{p^{-t}} \right)^{p^{-1}} x^j = \sum_{p^{t+1}j \leq n} c_{p^{t+1}j}^{p^{-(t+1)}} x^j.$$

This proves the induction. The last statement follows because $\lfloor n/p^t \rfloor = 0$ when $p^t > n$. $\square$

Theorem 3.2 (Index-chain conjugacy and functional components). *Let*

$$\mathcal{U}_n = \{u \in \{1, \dots, n\} : p \nmid u\}, \quad \ell_u = \max\{v \geq 0 : up^v \leq n\} \quad (u \in \mathcal{U}_n),$$

and let $S_u : \mathbb{F}_q^{\ell_u+1} \rightarrow \mathbb{F}_q^{\ell_u+1}$ be the nilpotent shift

$$S_u(z_0, \dots, z_{\ell_u}) = (z_1, \dots, z_{\ell_u}, 0).$$

For $f = \sum_{j=0}^n c_j x^j$, set

$$(5) \quad d_{u,v} = \sigma^{-v}(c_{up^v}), \quad \Psi(f) = (c_0, (d_{u,v})_{u \in \mathcal{U}_n, 0 \leq v \leq \ell_u}).$$

Then Ψ is an $\mathbb{F}_p$ -linear bijection from $X_{q,n}$ to

$$\mathbb{F}_q \times V_+, \quad V_+ = \prod_{u \in \mathcal{U}_n} \mathbb{F}_q^{\ell_u+1},$$

and, with $N = \prod_{u \in \mathcal{U}_n} S_u$,

$$(6) \quad \boxed{\Psi C \Psi^{-1} = \sigma^{-1} \times N.}$$

For $n = 0$, the product V_+ is the one-point zero space.

Consequently the directed functional graph has the following exact component interpretation. If $\mathcal{O}$ is an inverse-Frobenius cycle of length d in $\mathbb{F}_q$, then $\mathcal{O} \times V_+$ is one weak component, its directed cycle is $\mathcal{O} \times \{0\}$, and its size is dq^n . These are all weak components. For each $y \in \mathcal{O}$, the in-tree attached at $(y, 0)$ has q^n vertices; all such trees, over all periodic points, are isomorphic to the nilpotent in-tree of N at zero after the root cycle edge is suppressed.

More precisely, put $h(z) = \min\{t \geq 0 : N^t z = 0\}$ and

$$\mathcal{T}_y = \{(\sigma^{h(z)} y, z) : z \in V_+\}.$$

The vertices of $\mathcal{T}_y$ whose first core entry is $(y, 0)$ at time t number

$$(7) \quad \#\mathcal{T}_y(t) = \begin{cases} 1, & t = 0, \\ q^{n - \lfloor n/p^t \rfloor} - q^{n - \lfloor n/p^{t-1} \rfloor}, & 1 \leq t \leq D_{p,n}, \\ 0, & t > D_{p,n}, \end{cases}$$

where $D_{p,0} = 0$ and $D_{p,n} = 1 + \lfloor \log_p n \rfloor$ for $n > 0$. Finally, a vertex (y, z) in product coordinates has indegree

$$(8) \quad \#(\sigma^{-1} \times N)^{-1}(y, z) = \begin{cases} q^{n - \lfloor n/p \rfloor}, & z \in \text{im } N, \\ 0, & z \notin \text{im } N. \end{cases}$$

Proof. Every positive index has a unique factorization $j = up^v$ with $p \nmid u$, so the coordinates in (5) use every positive coefficient exactly once. Each power of Frobenius is an $\mathbb{F}_p$ -linear automorphism, and the inverse coordinate change is $c_{up^v} = \sigma^v(d_{u,v})$; hence Ψ is an $\mathbb{F}_p$ -linear bijection.

Write primes for coordinates after one application of $\mathcal{C}$. If $v < \ell_u$, then

$$d'_{u,v} = \sigma^{-v}(c'_{up^v}) = \sigma^{-v}(\sigma^{-1}(c_{up^{v+1}})) = \sigma^{-(v+1)}(c_{up^{v+1}}) = d_{u,v+1},$$

while $d'_{u,\ell_u} = 0$. The constant coordinate becomes $\sigma^{-1}(c_0)$. This proves (6); in particular, N is a finite product of nilpotent shifts.

Every $(x, z) \in \mathcal{O} \times V_+$ eventually reaches $\mathcal{O} \times \{0\}$, and no directed or reverse edge can change the inverse-Frobenius cycle containing its first coordinate. Thus $\mathcal{O} \times V_+$ is exactly one weak component and has size dq^n . For $z \neq 0$, nilpotence gives $h(Nz) = h(z) - 1$, and hence

$$(\sigma^{-1} \times N)(\sigma^{h(z)} y, z) = (\sigma^{h(z)-1} y, Nz) \in \mathcal{T}_y.$$

At $z = 0$ the outgoing edge is the cycle edge. Therefore $z \mapsto (\sigma^{h(z)} y, z)$ identifies the rooted in-tree of N with the tree attached at $(y, 0)$, and the sets $\mathcal{T}_y$ partition the component.

The rank of N^t over $\mathbb{F}_q$ equals the number of positive indices divisible by p^t , namely $\lfloor n/p^t \rfloor$. Thus $\#\ker N^t = q^{n - \lfloor n/p^t \rfloor}$, and differencing consecutive kernels proves (7). Finally, $Nw = z$ has no solution outside $\text{im } N$ and is a coset of $\ker N$ inside the image. Since $\#\ker N = q^{n - \lfloor n/p \rfloor}$ and inverse Frobenius has one preimage at the first coordinate, (8) follows. $\square$

Theorem 3.3 (Complete image and fibre census). *Let $t \geq 0$ and put $r_t = \lfloor n/p^t \rfloor$. Under the natural padded embedding $X_{q,r_t} \subseteq X_{q,n}$,*

$$(9) \quad \text{im}(\mathcal{C}^t) = X_{q,r_t}, \quad \#\text{im}(\mathcal{C}^t) = q^{r_t+1}.$$

For every target $g \in X_{q,n}$,

$$(10) \quad \#(\mathcal{C}^t)^{-1}(g) = \begin{cases} q^{n-r_t}, & \deg g \leq r_t, \\ 0, & \deg g > r_t. \end{cases}$$

Thus all and only targets in (9) have nonempty fibres, and all nonempty fibres are uniform. At $t = 0$ they are singletons; if $p^t > n$, there are q constant targets and every nonempty fibre has size q^n .

Proof. By theorem 3.1, the output coordinates are arbitrary precisely in positions $0, \dots, r_t$: each prescribed output has a unique preimage coefficient at each index $p^t j$, because inverse Frobenius is bijective. The remaining $n - r_t$ input coordinates are unobserved and may be chosen freely. This proves both cases of (10), including the empty fibre outside the displayed image. $\square$

The following complementary reconstruction starts from the already proved iterate formula rather than supplying a logically independent derivation of the dynamics. It packages that formula as finite-field linear algebra and recounts the fibre sizes and depth CDF.

Proposition 3.4 ($\mathbb{F}_p$ -linear rank factorization). *View $X_{q,n} \cong \mathbb{F}_q^{n+1}$ as an $\mathbb{F}_p$ -space. Let*

$$R_t(c_0, \dots, c_n) = (c_0, c_{p^t}, \dots, c_{r_t p^t}),$$

let Φ_t apply σ^{-t} coordinatewise, and let J_t pad with zeros. Then

$$(11) \quad \mathcal{C}^t = J_t \Phi_t R_t, \quad \text{rank}_{\mathbb{F}_p}(\mathcal{C}^t) = a(r_t + 1), \quad \dim_{\mathbb{F}_p} \ker(\mathcal{C}^t) = a(n - r_t).$$

Moreover,

$$(12) \quad \#\{f : \mathcal{C}^t(f) \in K\} = q^{n+1-r_t}.$$

Complementary route II: ranks. The restriction R_t is an $\mathbb{F}_q$ -linear surjection, Φ_t is an $\mathbb{F}_p$ -linear automorphism, and J_t is an injective $\mathbb{F}_q$ -linear map. Their factorization follows directly from the displayed coordinates, and rank-nullity gives (11). Hence every nonempty fibre is an affine coset of a kernel of size q^{n-r_t} , recounting (10). Finally, $\mathcal{C}^t(f)$ is constant exactly when the r_t selected positive coordinates vanish. These are r_t independent $\mathbb{F}_q$ constraints, which proves (12). $\square$

4. CORE-ENTRY DISTRIBUTION AND SHARP DEPTH

For an integer $j > 0$, let $v_p(j)$ be its p -adic valuation. An occupied coefficient at j moves down its p -chain until the first index not divisible by p , then disappears at the next step.

Theorem 4.1 (Every depth shell and the deepest shell). *If f is constant, then $\tau(f) = 0$. If f is nonconstant, then*

$$(13) \quad \tau(f) = 1 + \max\{v_p(j) : 1 \leq j \leq n, c_j \neq 0\}.$$

For every $t \geq 0$,

$$(14) \quad \#\{f : \tau(f) \leq t\} = q^{n+1-\lfloor n/p^t \rfloor}, \quad \mathbb{P}_f \text{ unif.}(\tau(f) \leq t) = q^{-\lfloor n/p^t \rfloor}.$$

When $n = 0$, all q states have depth zero and the maximum depth is $D_{p,0} = 0$. When $n \geq 1$, set

$$(15) \quad D_{p,n} = 1 + \lfloor \log_p n \rfloor, \quad s_{p,n} = \left\lfloor \frac{n}{p^{D_{p,n}-1}} \right\rfloor \in \{1, \dots, p-1\}.$$

Then $D_{p,n}$ is the sharp maximum of τ , and

$$(16) \quad \#\{f : \tau(f) = t\} = q^{n+1-\lfloor n/p^t \rfloor} - q^{n+1-\lfloor n/p^{t-1} \rfloor}, \quad 1 \leq t \leq D_{p,n},$$

$$(17) \quad \#\{f : \tau(f) = D_{p,n}\} = q^{n+1} - q^{n+1-s_{p,n}}.$$

There are no states at larger depth.

Coefficient-forest proof. Fix an occupied index $j > 0$ and write $j = p^v u$ with $p \nmid u$. At time $s \leq v$, (4) carries its nonzero coefficient to the positive index $p^{v-s} u$; inverse Frobenius cannot turn a nonzero element into zero. At time $v+1$ that coefficient is not selected. Taking the last disappearance time over all occupied positive indices proves (13); the constant case is separated because its occupied positive-index set is empty.

The event $\tau(f) \leq t$ says exactly that every positive coefficient whose index is divisible by p^t vanishes. There are $\lfloor n/p^t \rfloor$ such constrained coordinates, so the other $n+1 - \lfloor n/p^t \rfloor$ coordinates are arbitrary. This proves (14); it also agrees with the complementary rank count (12). Subtracting consecutive CDF values gives (16).

For $n \geq 1$, the largest valuation among $1, \dots, n$ is $D_{p,n} - 1$, since $p^{D_{p,n}-1} \leq n < p^{D_{p,n}}$. Thus the maximum is sharp, witnessed by $x^{p^{D_{p,n}-1}}$. Exactly $s_{p,n}$ indices in $[1, n]$ are divisible by $p^{D_{p,n}-1}$. A state is deepest precisely when at least one of those coefficients is nonzero. Subtracting the $q^{n+1-s_{p,n}}$ states in which all of them vanish from the full phase space proves (17) and completes all boundary cases. $\square$

5. INVERSE-FROBENIUS CORE, CYCLES, AND ZETA

Let μ denote the number-theoretic Möbius function. The transient coordinates disappear, but the constant coefficient continues to move.

Theorem 5.1 (Complete periodic census). *For every $n \geq 0$,*

$$(18) \quad \text{Core}(\mathcal{C}) = \text{Per}(\mathcal{C}) = K, \quad \mathcal{C}|_K = \sigma^{-1}.$$

For every $m \geq 1$,

$$(19) \quad \boxed{\# \text{Fix}(\mathcal{C}^m) = p^{\gcd(a,m)}}.$$

For each divisor d of a , define

$$(20) \quad A_d = \sum_{e|d} \mu(d/e) p^e.$$

Then A_d is the number of points of exact period d , and A_d/d is the number of d -cycles. No period outside the divisors of a occurs. The full functional graph therefore has A_d/d weak components based on d -cycles, each with dq^n vertices and the attached trees described in theorem 3.2. The Artin–Mazur zeta function of the full finite map is the formal series

$$(21) \quad \begin{aligned} \zeta_{\mathcal{C}}(z) &:= \exp \left(\sum_{m \geq 1} \# \text{Fix}(\mathcal{C}^m) \frac{z^m}{m} \right) \\ &= \boxed{\prod_{d|a} (1 - z^d)^{-A_d/d}}. \end{aligned}$$

These statements include $a = 1$, where all $q = p$ constants are fixed and $\zeta_{\mathcal{C}}(z) = (1 - z)^{-p}$, and $n = 0$, where the phase space already equals the core.

Stable-image and subfield proof. By (9), the images decrease to K , proving the first equality in (18) for the stable image. If f is periodic, say $\mathcal{C}^m(f) = f$, then $f \in \text{im}(\mathcal{C}^{km})$ for every k , so $f \in K$. Conversely, (4) sends a constant c to $\sigma^{-1}(c)$, an automorphism whose order divides a ; hence every constant is periodic.

A fixed point of $\mathcal{C}^m$ is therefore a constant fixed by σ^{-m} , equivalently by σ^m . To count these constants directly, note that the elements of $\mathbb{F}_{p^a}$ are the roots of $T^{p^a} - T$, while the σ^m -fixed elements are also roots of $T^{p^m} - T$. The Euclidean algorithm on the exponents gives

$$\gcd(T^{p^a} - T, T^{p^m} - T) = T^{p^{\gcd(a,m)}} - T.$$

This polynomial is separable and has degree $p^{\gcd(a,m)}$, proving (19) without an uncounted field-theory step. Every orbit length divides a . For $d | a$, the p^d elements fixed by the d th iterate are partitioned by their exact periods, so $p^d = \sum_{e|d} A_e$. Divisor Möbius inversion gives (20), and division by d counts cycles.

Finally, a cycle of length d contributes $\exp(\sum_{k \geq 1} z^{kd}/k) = (1 - z^d)^{-1}$ to the zeta function. Multiplying over the A_d/d cycles of each possible length proves (21). Transient trees contribute no fixed points and hence no zeta factor. $\square$

6. AN EXACT LATTICE-DEPTH LIMIT

The sharp depth changes when n crosses powers of p . Fixing the position inside one such lattice interval removes that oscillation.

Theorem 6.1 (Layerwise stabilization and limiting defect). *Fix p , a , and a real number $1 \leq \alpha < p$. For integers $L \geq 0$, let*

$$n_L = \lfloor \alpha p^L \rfloor, \quad F_L \sim \text{Unif}(X_{q, n_L}), \quad \Delta_L = L + 1 - \tau(F_L).$$

Then the sharp depth at degree bound n_L is exactly $L + 1$, so $\Delta_L \in \{0, \dots, L + 1\}$. For every fixed $k \geq 1$ and every $L \geq k - 1$,

$$(22) \quad \boxed{\mathbb{P}(\Delta_L \geq k) = q^{-\lfloor \alpha p^{k-1} \rfloor}}.$$

Consequently Δ_L converges in distribution to a nonnegative integer-valued variable Δ with tails

$$(23) \quad \mathbb{P}(\Delta \geq 0) = 1, \quad \mathbb{P}(\Delta \geq k) = q^{-\lfloor \alpha p^{k-1} \rfloor}, \quad k \geq 1,$$

or, equivalently,

$$(24) \quad \mathbb{P}(\Delta = r) = S_r - S_{r+1}, \quad S_0 = 1, \quad S_r = q^{-\lfloor \alpha p^{r-1} \rfloor} \quad (r \geq 1).$$

Thus the asserted “limit” is stronger than asymptotic convergence: every fixed tail is eventually exact.

Proof. The inequalities $p^L \leq n_L < p^{L+1}$ follow from $1 \leq \alpha < p$, including the endpoint $\alpha = 1$. Hence (15) gives maximum depth $L + 1$. If $L \geq k - 1$, then by (14),

$$\begin{aligned} \mathbb{P}(\Delta_L \geq k) &= \mathbb{P}(\tau(F_L) \leq L + 1 - k) \\ &= q^{-\lfloor n_L / p^{L+1-k} \rfloor}. \end{aligned}$$

For an integer $M \geq 1$, the identity $\lfloor \lfloor x \rfloor / M \rfloor = \lfloor x / M \rfloor$ gives

$$\left\lfloor \frac{n_L}{p^{L+1-k}} \right\rfloor = \lfloor \alpha p^{k-1} \rfloor,$$

which proves (22). The sequence S_r decreases to zero, so the differences in (24) are nonnegative and sum to one. Stabilization of every tail proves convergence in distribution. $\square$

Remark 6.2 (Why the lattice window is explicit). The restriction $\alpha < p$ keeps n_L in the depth interval $[p^L, p^{L+1})$. At $\alpha = p$ the maximum depth shifts to $L + 2$; that endpoint is represented instead by the allowed choice $\alpha = 1$ after incrementing L . No assertion is made for $k > L + 1$ before stabilization, when the finite variable has no such tail.

7. TEMPORAL RECOVERY OF THE PARAMETERS

The periodic sequence alone loses the degree bound, while phase size alone cannot separate the field size from the number of coefficients. Together they are rigid.

Theorem 7.1 (Temporal signature rigidity). *Let*

$$N = |X_{q,n}|, \quad F_m = \# \text{Fix}(\mathcal{C}^m) \quad (m \geq 1).$$

The signature $(N, (F_m)_{m \geq 1})$ determines (p, a, n) by

$$(25) \quad \boxed{p = F_1, \quad q = \max_{m \geq 1} F_m, \quad a = \min\{m \geq 1 : F_m = q\}, \quad n = \log_q N - 1.}$$

In particular, two bounded Cartier systems with equal signatures have equal parameter triples. The formula remains valid for $a = 1$ and for $n = 0$. Here $(F_m)_{m \geq 1}$ means the full infinite fixed-count sequence; no claim is made that an unspecified preassigned finite prefix is sufficient.

Proof. Equation (19) gives $F_1 = p$. It also gives $F_m \leq p^a = q$, with equality exactly when $a \mid m$. Thus the maximum is q and its first occurrence is $m = a$. Finally, $N = q^{n+1}$, so the last formula in (25) follows. When $a = 1$, the maximum occurs first at one; when $n = 0$, $N = q$ and the recovered degree bound is zero. $\square$

8. EXACT CONTROLS AND SCOPE

The accompanying standard-library verifier constructs the polynomial-basis fields $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_4, \mathbb{F}_8, \mathbb{F}_9$, and $\mathbb{F}_{16}$ literally. It exhausts the main lanes through degree bounds 7, 5, 5, 4, 3, 2, respectively, as well as the $n = 0$ boundary for every field. It compares direct orbit iteration with (4), verifies the forward and inverse chain-coordinate maps and (6) state by state, enumerates every image target and fibre, checks every depth shell, per-root entry layer, component size, and deepest shell, reconstructs core cycles and fixed counts, compares zeta coefficients from fixed counts and from the Euler product, and tests 33 rational lattice lanes through $L = 9$. The fresh run recorded **2,259,162 exact assertion executions; PASS**. This raw counter increments once per executed check, including looped coordinate checks and aggregate identities; it is not a count of logically independent claims.

The control also rejects three tempting false simplifications. For $a > 1$, inverse Frobenius makes $\mathcal{C}$ generally $\mathbb{F}_p$ -linear but not $\mathbb{F}_q$ -linear. Equal-degree polynomials can have different depths because lower occupied indices matter. Extension fields have nontrivial core cycles, so replacing (21) by $(1 - z)^{-q}$ is wrong unless $a = 1$. These finite checks are falsification evidence, not proofs of the general theorems and not novelty evidence.

The residual claim is limited to the exact bounded Cartier specialization proved above—its coefficient, fibre, temporal, and component formulas— together with the lattice stabilization and parameter-recovery conjunction. Direct Cartier extraction, Frobenius/subfield structure, restriction-of- scalars linearization, cyclic-nilpotent product theory, attached-tree machinery, rank/kernel counts, Möbius cycle conversion, and generic finite linear functional graphs are subtracted with zero credit. A specialist direct-owner search remains necessary before any external use. External release, novelty, and priority remain on **HOLD**.

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